

Tender Ref: CPWD/WSD/2026/EL-049

Date of Issuance: 14-September-2026

Item: Pumping Machinery and Auxiliary Motor Electrification Works

SECTION IV: TECHNICAL SPECIFICATIONS & STANDARDS OF WORK

1. SCOPE OF SUPPLY:

The contractor shall supply, install, test and commission Three-Phase Squirrel Cage Induction Motors for raw water pumping station duty.

2. TECHNICAL SPECIFICATIONS:

- Motor Capacity: 15 kW (20 HP continuous S1 duty)
- Electrical System: 415 V $\pm$ 10%, 50 Hz $\pm$ 5%, 3-Phase AC supply
- Operating Speed: 1440 RPM (4 Pole)
- Enclosure & Ingress Protection: Totally Enclosed Fan Cooled (TEFC) with minimum IP55 protection suitable for humid pumphouse conditions.
- Energy Efficiency: The motor must conform to Premium Energy Efficiency Class (IE3) in accordance with applicable Indian Standards.
- Insulation: Class F insulation with temperature rise limited to Class B limits.

3. GOVERNING CODES & STANDARDS:

- Induction motor performance and energy efficiency must strictly conform to IS 12615.
- Note for bidders: Tender specification originally referred to legacy standard IS 325:1996 which has been audited.
- Testing for efficiency and losses shall be conducted in accordance with IS 15999 (Part 2/Sec 1).
- Degrees of protection by integral enclosure design shall conform to IS/IEC 60034-5.
- Terminal markings and phase rotation shall comply with IS/IEC 60034-8.
- Earthing pit and body earthing shall be provided as per Code of Practice IS 3043:2018.

4. CERTIFICATION REQUIREMENTS:

- The equipment must carry mandatory BIS Scheme-I (ISI Mark) under the Electrical Motors Quality Control Order (QCO).
- Manufacturer test certificate for routine tests and type tests must be submitted along with delivery.